

VENKAT DINESH PASUPULETI

Machine Learning Engineer · GenAI & LLM Systems · RAG Architecture · MLOps · Fraud & Risk AI

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Portfolio: venkatdinesh.vercel.app · [Linkedin](#) · [GitHub](#) · [Live RAG Demo: venkatdinesh-rag.hf.space](#)

7+ Years ML/AI · 90% Fraud Precision · ~\$500K/yr Savings · 557 Production APIs · Live Production RAG · PySpark on Databricks · Open Work Permit

PROFESSIONAL SUMMARY

Machine Learning Engineer with 7+ years building production AI systems across financial services and enterprise SaaS - from gradient-boosted fraud models saving ~\$500K/yr under RBI regulatory oversight, to LLM-powered agentic platforms serving enterprise clients at <2s latency. Deep, hands-on expertise across the full modern AI and data stack: classical ML (XGBoost, LightGBM, SHAP, PSI drift monitoring), GenAI & LLM systems (Anthropic Claude, OpenAI GPT-4.1, production RAG with hybrid retrieval and CI-gated evaluation), large-scale data engineering (PySpark on Databricks, ETL across 5+ sources, 1M+ row pipelines), and MLOps (Azure ML, GKE/Kubernetes, Docker, MLflow, GitHub Actions). SHAP explainability documentation adopted as RBI regulatory audit artefacts; 365-day audit trails with 7-year retention; hallucination prevention engineered into production prompts. Valid Canadian open work permit - eligible for any employer immediately, no sponsorship required.

CORE TECHNICAL SKILLS

Machine Learning: XGBoost, LightGBM, CatBoost, Extra Trees, Scikit-learn, feature engineering, classification, regression, clustering, propensity modelling, anomaly detection, A/B testing, statistical significance

GenAI & LLM Systems: Anthropic Claude (Sonnet 4.5), OpenAI GPT-4.1, RAG (hybrid BM25+FAISS, RRF), cross-encoder reranking, embeddings, prompt engineering, structured extraction, zero-shot classification, LLM evaluation

Agentic AI: custom tool schemas (8 in production), agent orchestration, intent classification, KB-conditional routing, hallucination prevention (negative-example anchoring), MCP-compatible patterns, multi-tenant AI platforms

Data Engineering: PySpark (Databricks), SQL (advanced), pandas, NumPy, ETL (5+ heterogeneous sources), 1M+ row pipelines, feature engineering at scale, data quality frameworks (61 to 89%), schema standardisation, OCR/unstructured data

Explainability & Governance: SHAP (force plots, summary plots), PSI drift monitoring, model documentation for regulators (RBI), RBAC (71 permissions), 7-yr audit logging, Responsible AI, hallucination guardrails

MLOps & Cloud: Azure ML (managed endpoints, scoring scripts, App Insights), GCP (GKE, Cloud Run), MLflow, Docker multi-stage, Kubernetes autoscaling, GitHub Actions CI/CD, model registries, automated retraining

Production Python: FastAPI (557 endpoints), Pydantic, async Python, REST/microservices, WebSocket APIs, Redis, unit testing, CI-gated deployment

Platforms & Tooling: Databricks (PySpark notebooks), Azure ML, Hugging Face Spaces, MongoDB aggregation, PostgreSQL, Power BI, Docker, Kubernetes, Git

Computer Vision & NLP: YOLOv3, Faster R-CNN, TensorFlow, OpenCV, 5-class sentiment pipelines, multilingual NLU (9 languages), voicemail detection, call-success scoring

PROFESSIONAL EXPERIENCE

Lead AI/ML Engineer · DeskIQ.ai · Jay Analytix Inc

Remote, Canada | Aug 2025 - Present

- Architected an agentic voice AI platform** on GPT-4.1 + Claude Sonnet 4.5 - 557 FastAPI endpoints across 42 production modules on GCP, serving enterprise clients with <2-second end-to-end latency on GKE autoscaling.
- Designed 8 custom tool schemas** with FastAPI execution endpoints and a zero-shot intent classification gate; engineered dual-prompt system (462-line inbound / 19-line outbound) with hallucination prevention via negative-example anchoring.
- Built analytics data pipeline on Databricks:** PySpark on Databricks notebooks to aggregate and transform client interaction data into structured reporting datasets; enabled enterprise client analytics dashboards with **40% faster reporting cycles**.
- Deployed production RAG system** (live: venkatdinesh-rag.hf.space) - hybrid BM25+FAISS retrieval, cross-encoder reranking, citation enforcement, streaming, per-stage observability, 50+ golden-set test suite, CI-gated evaluation; \$0/month infra.
- Implemented enterprise governance:** 71-permission RBAC middleware, KB-conditional routing, 365-day audit logging with 7-year retention, GitHub Actions CI/CD.
- Built NLP pipelines:** 5-class sentiment, voicemail detection, call-success scoring, spoken-email parsing; 9-language multilingual NLU; post-call analytics pipeline (Twilio webhook to MongoDB to insights dashboards).
- Delivered 12+ enterprise integrations** (HubSpot, Zapier, Calendly, Google Calendar, SendGrid, Twilio, Stripe) with Redis session management and WebSocket APIs; deployed via GitHub Actions CI/CD; 90% booking automation, 60% error reduction.

AI/ML Engineer (Contract) · DeskIQ.ai · Jay Analytix Inc

Remote, Canada | Feb 2025 - Aug 2025

- Delivered first production feature (outbound dialler prompt engineering) within 2 weeks; built initial RAG chatbot integration and agentic tool schemas. **Promoted to Lead AI/ML Engineer after 6 months.**

Lead Data Scientist · Vreedhi Financial Services · Dr. Reddy's Foundation

Hyderabad, India | Jun 2023 - Jan 2025

- PySpark ETL and feature engineering on Databricks:** built large-scale data transformation pipelines on Databricks using PySpark across mixed loan application, transaction, and customer data (1M+ rows); cleaned, deduped, and engineered features feeding production ML models.
- Deployed production fraud detection:** XGBoost classifier on Azure ML (managed endpoint + scoring script) achieving **90% precision**; automated triage saved **~\$500K/yr** in analyst overhead within an RBI-regulated environment.
- Caught silent model drift in production** via PSI monitoring (90% to 74% precision degradation); root-caused to population distribution shift, retrained on refreshed rolling-window data, restored precision **within one sprint - zero SLA breach**; deployed automated retraining trigger.
- Built document intelligence pipeline:** Azure Form Recognizer + custom OCR for loan document extraction; processing 2 days to 4 hours, **50% manual review reduction**.
- Pioneered explainable AI for compliance:** SHAP force plots adopted as **RBI regulatory audit artefacts**; propensity model (XGBoost ~70% acc.) + segmentation (Extra Trees 82% acc.) on 100K+ records - ~20% engagement uplift.
- Recognition:** Annual Best Performance Award 2024; 3-Year Service Award 2024.

Data Scientist · Vreedhi Financial Services · Dr. Reddy's Foundation

Hyderabad, India | May 2021 - May 2023

- ML pipelines:** feature engineering to Azure ML deployment for credit risk scoring; **35% underwriting workload reduction**. Python + SQL ETL across 5+ sources; data quality 61% to 89% via deduplication, outlier handling, and schema standardisation.
- Computer vision:** YOLOv3 + Faster R-CNN (TensorFlow + OpenCV) for document image classification in loan processing pipeline.
- Mentored 3 junior data analysts;** introduced code-review practices and standardised Jupyter to Python module migration for reproducibility.

Data Analyst · Vinfosoft Technologies

Hyderabad, India | Dec 2019 - Apr 2021

- Designed Power BI + Python BI dashboards** for retail banking clients; automated weekly reporting pipeline saving **12 hrs/week**; SQL ETL query optimisation cut report generation time by 60%.
- Delivered A/B test analysis framework** adopted across 4 product teams - first statistical significance testing in the organisation.

Analyst · Vinfosoft Technologies

Hyderabad, India | Dec 2018 - Dec 2019

- SQL data warehouse pipelines for 3 retail banking clients; reusable ETL modules cut onboarding 40%; executive KPI dashboards reduced ad-hoc requests by 50%.

KEY PROJECTS

Production RAG System · [Live: venkatesh-rag.hf.space](https://venkatesh-rag.hf.space) · [GitHub](#)

- Hybrid BM25+FAISS with RRF; cross-encoder reranking (top_k=5); filter_score=0.7 hallucination guard; citation enforcement; streaming; per-stage latency/token/confidence observability; 50+ golden-set tests; CI gates on retrieval recall + faithfulness; Docker multi-stage. \$0/month.

DeskIQ Agentic Voice AI Platform (Production)

- Multi-tenant AI phone platform: 557 FastAPI endpoints, 42 modules, 8 tool schemas, dual-prompt engineering, zero-shot intent classification, 71-permission RBAC, 365-day audit logging, GKE autoscaling - live in production serving enterprise clients.

AI Portfolio with Claude-Powered Chat · venkatesh.vercel.app

- Full-stack HTML/CSS/JS with Anthropic Claude API chat assistant via secure Vercel serverless proxy - visitors can interrogate my experience conversationally. AI-first development in practice.

Fraud Detection & Model Drift Recovery (Vreedhi - Production)

- PySpark on Databricks for 1M+ row ETL and feature engineering - XGBoost at 90% precision on Azure ML - PSI drift detection - sprint-cycle retraining - automated retraining trigger - SHAP documentation adopted by RBI auditors. Full MLOps lifecycle in regulated production.

EDUCATION

Post-Graduate Diploma - Data Analytics

St. Clair College, Windsor, ON, Canada | GPA 3.25/4.0 | Jan 2025 - Apr 2026

Coursework: Machine Learning, Deep Learning, Big Data Analytics, Cloud Computing, Statistical Modelling, Data Visualisation

Post-Graduate Diploma - Data Science (Grade A)

INSOFE (International School of Engineering), India | 2020-2021

Hackathon: Ranked 8/120 · Coursework: Applied ML, NLP, Neural Networks, Feature Engineering, Model Deployment, Statistical Inference

Bachelor of Engineering - Electronics & Communication

SRIT, India | 2012-2016

CERTIFICATIONS

AI Engineer for Developers Associate - DataCamp, Jun 2026 (exp Jun 2028) · AIEDA0015980374013
Prompt Engineering & Programming with OpenAI - Columbia University (Columbia+), Jan 2026 · 172702485
Large Language Models Specialisation - Google Cloud / Coursera, Dec 2025 · ZHVPKDL7VIVR
Introduction to Responsible AI - Google Cloud / Coursera, Dec 2025 · QUZ4VM7F2WZQ
Microsoft Azure Data Scientist Associate (DP-100) - Microsoft, Aug 2023 - applied in production at Vreedhi Financial Services · D349422484A448DF

AWARDS & RECOGNITION

- **Annual Best Performance Award 2024** - Vreedhi Financial Services / Dr. Reddy's Foundation
- **3-Year Service Award 2024** - Dr. Reddy's Foundation
- **Hackathon Rank 8/120** - INSOFE Data Science Hackathon
- **Promoted to Lead AI/ML Engineer within 6 months** - DeskIQ.ai / Jay Analytix Inc

CAREER HIGHLIGHTS AT A GLANCE

- **Scale:** 557 FastAPI endpoints · 42 production modules · 12+ enterprise integrations · 71 RBAC permissions · 9-language NLU · 1M+ rows PySpark pipelines · 100K+ records modelled
- **Business impact:** ~\$500K/yr fraud savings · 50% OCR reduction · 35% underwriting reduction · ~20% engagement uplift · 90% booking automation · 40% reporting acceleration
- **Engineering rigour:** zero SLA breach on drift recovery · CI-gated RAG evaluation · 50+ test suites · RBI-adopted explainability documentation · 7-year audit retention
- **Availability:** Valid Canadian open work permit - any employer, immediate start, no sponsorship required. Open to hybrid (GTA) or remote.